

9.2 Writing a Quadratic Function From a Graph

Lesson Introduction

 Benchmarks	MA.912.AR.1.2 Re-arrange equations or formulas to isolate a quantity of interest.		 Learning Targets	- I can write a quadratic function to represent the relationship between two quantities from a graph. - I can re-write a quadratic equation from vertex form to standard form. - I can solve for a variable of interest.
	MA.912.AR.3.4 Write a quadratic function to represent the relationship between two quantities from a graph, a written description, or a table of values within a mathematical or real-world context.			
 Benchmark Coherence	Building On MA.8.AR.2.1 Solve multi-step linear equations in one variable, with rational number coefficients. Include equations with variables on both sides.		Working Toward MA.912.AR.3.4 Write a quadratic function to represent the relationship between two quantities from a graph, a written description, or a table of values within a mathematical or real-world context clarification 3.	
 Essential Questions	- Which key feature(s) of a parabola can help determine a quadratic function from a graph? - Which form of a quadratic is easiest to use when writing the equation from a graph? Why? - How can you use key features from a graph to write the equation of the quadratic?			
 Lesson Narrative	Students will extend their understanding of writing quadratic equations from given points to a given graph. Students will apply their knowledge of quadratic functions to real-world scenarios to write the function both from a graph and when given a vertex and a point from a graph.			
 Component Summary	9.2.1 Warm-Up (~5 min.)	Gymnastics vaulting exploration (<i>SE p. 65</i>)	9.2.3 Collaboration (15 min.)	Collaboration on writing a quadratic function given a vertex and a point from the graph (<i>SE p. 67</i>)
	9.2.2 Guided Instruction (10 min.)	Writing a quadratic function in vertex form given a graph (<i>SE p. 66</i>)	9.2.4 Your Turn! (10 min.)	Independent practice with writing a quadratic function in vertex and standard form when given a graph, vertex, and a point on the graph (<i>SE p. 69</i>)
	9.2.5 Wrap-Up (~5 min.)	Self-reflection on their level of confidence with writing equations of quadratic functions (<i>SE p. 70</i>)		
 Resources	Glossary	Materials	Additional Preparation	
	<u>Key Terms:</u> N/A <u>Additional Glossary:</u> x-intercept - Vertex form - Standard form - Equidistant	N/A	N/A	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may incorrectly square a binomial. For example, students may simplify $(x + 1)^2 = x^2 + 1$. - Students may assume a quadratic with a maximum means positive leading coefficient, while a quadratic with a minimum has a negative leading coefficient.	- In 9.2.3 Collaboration, consider permitting students to follow the setup in the activity with their partners without explicit instruction. - In 9.2.3 Collaboration, question 3 and 9.2.4 Your Turn, question 3, the a value is a fraction. When students are asked to convert to standard form, some students may struggle with the fraction operations. Consider checking for their conceptual understanding of the conversion to determine whether intervention on the fraction operations or the conversion is the root issue. Allow students to use decimals in place of fractions.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Take Turns 9.2.3 Collaboration provides students with a collaborative opportunity to work with a partner on writing a quadratic function using a vertex and point from a graph. The discussion questions within this activity provide students with the opportunity to discuss their answer, process, and observations to allow the others to share theirs to come to a consensus.	MA.K12.MTR.4.1: Discussion and Reflection 9.2.3 Collaboration provides students a collaborative opportunity to engage in discussions about writing a function using a vertex and point from a graph. Students will articulate their mathematical thinking and engage in dialogue about others' mathematical thinking.
 Differentiate Instruction	Leverage digital graphing utility to provide additional entry points and representations for the graphs of quadratic equations. Testing points on a parabola algebraically may make sense for some students, while other students may benefit from being able to explore and visually connect solutions to their representation on a graph. For example, highlighting the points in question 2 during Guided Instruction might be the catalyst for students to make connections between the key features and their algebraic representation.	
Lesson Notes:		

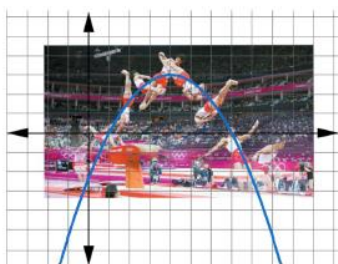
9.2.1 Warm-Up: Vaulting Moves

Component Overview: Students will apply what they have learned about the “ a ” value and x -intercepts to a real-world example involving vaulting moves.



9.2.1 Warm-Up: Vaulting Moves

Vaulting is an exercise in which gymnasts leap over an apparatus and perform a skill in mid-air. The image shows a frame-by-frame pictorial of a gymnast’s vault routine. Based on the compilation of moves performed by the gymnast, a sketch of a parabola was drawn.



1. Explain whether the a value of the parabola is positive or negative.
Answers will vary. The a value of the parabola would be negative because the parabola opens downward.

2. If each square is 1 by 1, what are the vertex and x -intercepts of the parabola?

vertex	(4,3)
x -intercepts	(1,0) and (7,0)



Consider finding a short video of someone completing a vault routine to give students an engaging entry point to make sense of real-world applications for quadratics as they engage in writing their equations throughout the lesson.



How can you tell, by looking at the graph, if the a -value is positive or negative?

What if you didn't have the graph, how could you tell if the a -value is positive or negative?

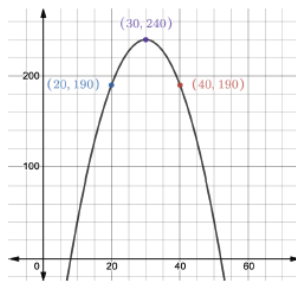
9.2.2 Guided Instruction: Writing a Quadratic Function in Vertex Form From a Graph

Component Overview: Students will receive guided instruction on how to write a quadratic function in vertex form from a graph.



9.2.2 Guided Instruction: Writing a Quadratic Function in Vertex Form from a Graph

Hudson is starting a lawn-mowing business in his neighborhood. He creates a graph to help him determine what to charge customers per mow to maximize his profits. He uses x to represent the cost of each lawn he mows and y to represent the profit in dollars (\$).



- The point ☐ (20, 190) ☒ (30, 240) ☐ (40, 190) represents the maximum profit

Hudson can make. This point on a graph is called the vertex of the parabola.

- Discuss with your partner what you notice about points (20, 190) and (40, 190).
Answers will vary. The points are the same distance from the vertex. The profit is \$190 when it costs Hudson \$20 or \$40 to mow a particular lawn.
- Points (20, 190) and (40, 190) are equidistant from the vertex (30, 240). Identify two other points on the graph that are equidistant from the vertex (30, 240).
Answers will vary. (10, 40) and (50, 40) or any other points on the parabola with different x-values and the same y-values.



Routines for Math Literacy

Consider leveraging word-problem reading strategies like the *Three Reads* routine to the problem before moving through the questions in this activity.

A *Notice and Wonder* routine may also act as a catalyst for making connections between key features on the graph and their algebraic representation.



Use inquiry to transfer visual observations to abstract representations, connecting what students did in Lesson 1 to this context of using graphs in Lesson 2.

- Why do you think a business has a maximum profit?
- Why might the y -value decrease as the x -increases after 30 in this scenario?
- What is one item that you know sold well for a few years but then sales dropped off?
- How could rising and falling sales affect a company's profit?
- Will every point on the parabola have another point that is also equidistant from the vertex? Why or why not?
- What is the axis of symmetry? What is the distance (20, 190) and (40, 190) are from the axis of symmetry?

9.2.2 Guided Instruction: Writing a Quadratic Function in Vertex Form From a Graph (continued)

4. To find the equation of the quadratic function

represented by the graph, it would be easiest to use

- ☐ factored form
- ☐ standard form
- ☒ vertex form

because the graph provides the

vertex and at least one other point.

The steps in the table show how to use the point (20, 190) and the vertex (30, 240) to find the quadratic function in vertex form $f(x) = a(x - h)^2 + k$, where (h, k) is the vertex.

Substitute values for the vertex (30, 240).	$f(x) = a(x - 30)^2 + 240$
Substitute values for the point (20, 190).	$190 = a(20 - 30)^2 + 240$
Solve for the variable a .	$190 = a(-10)^2 + 240$ $190 - 240 = a(100)$ $-50 = a(100)$ $a = -\frac{1}{2}$
Substitute in the vertex and a value.	$f(x) = -\frac{1}{2}(x - 30)^2 + 240$

The quadratic equation Hudson graphed is $y = -\frac{1}{2}(x - 30)^2 + 240$.



Students may make mistakes when inputting the values into the function. Consider using colored writing utensils such as highlighters, markers, etc. to simulate a similar color scheme as the book does to draw attention to which values were put into which spots. For auditory learners, continue to verbalize what numbers were put where and why.



In Lesson 1, students used factored form to write the equation of the quadratic. Engage students throughout the lesson in understanding why vertex form may be the best form when writing a quadratic from a graph, but that x -intercept could also be used if the x -intercepts are known.



Why is vertex form the best form for writing a quadratic from a graph? Could x -intercept form also be used if the x -intercepts are known?

9.2.3 Collaboration: Writing a Quadratic Function Using a Vertex and Point From a Graph

Component Overview: Students will work with a partner to collaborate on writing a quadratic function given a vertex and a point from the graph. Introduce students to the context of question 1 and how the scenario is the same from the Guided Instruction activity. Allow students to work on all three questions with their partner or group.



9.2.3 Collaboration: Writing a Quadratic Function Using a Vertex and Point from a Graph

1. With your partner, use the other point identified on Hudson's graph, (40, 190), along with the same vertex, (30, 240), to generate another quadratic equation in vertex form.

Substitute values for the vertex (30, 240).	$f(x) = a(x - 30)^2 + 240$
Substitute values for the point (40, 190).	$190 = a(40 - 30)^2 + 240$
Solve for the variable a .	$190 = a(10)^2 + 240$ $190 - 240 = a(10)^2$ $-50 = a(100)$ $a = -\frac{1}{2}$
Substitute in the vertex and a value.	$f(x) = -\frac{1}{2}(x - 30)^2 + 240$

- a. The quadratic equation with vertex (30, 240) and point (40, 190) is $y = -\frac{1}{2}(x - 30)^2 + 240$.
- b. Discuss with your partner what you notice about quadratic functions with the same vertex and two points that are equidistant from the vertex. Summarize your discussion.

Answers will vary. The equations in vertex forms are the same. You could use either point when you are finding the vertex form of a quadratic.



Will a different point on the same parabola always produce the same quadratic equation?



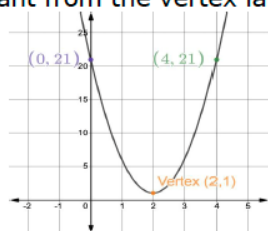
For a challenge, ask students to determine how the a -value can be used as a slope to find a symmetrical pair of points by rise over run from the vertex as one slope. Follow up by asking if another pair of points could be found using the same "rise over run."



Listen for potential misconceptions about not being the same. Allow students to share the discussions that they had with their partners for the whole class to hear. Relate points that are equidistant with the axis of symmetry.

9.2.3 Collaboration: Writing a Quadratic Function Using a Vertex and Point From a Graph (continued)

2. A quadratic function is graphed with the vertex and two points equidistant from the vertex labeled.



- a. Write the equation in vertex form for the quadratic function graphed.

Substitute values for the vertex (2, 1).	$y = a(x - 2)^2 + 1$
Substitute values for the point (0, 21) or (4, 21).	$21 = a(0 - 2)^2 + 1$
Solve for the variable a .	$21 = a(0 - 2)^2 + 1$ $21 = a(-2)^2 + 1$ $21 - 1 = a(-2)^2$ $20 = a(4)$ $20 = a(4)$ $a = 5$
Substitute in the vertex and a value.	$y = 5(x - 2)^2 + 1$



*How do you know that (0, 21) and (4, 21) are equidistant?
How can you use the axis of symmetry to show that they are equidistant?*

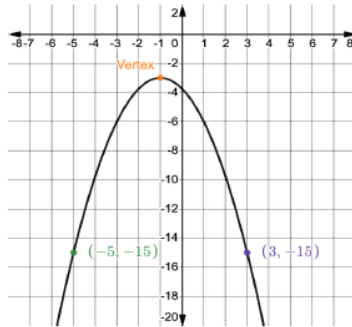
- b. The quadratic equation is $y = 5(x - 2)^2 + 1$.

- c. Rewrite the equation into standard form

$$\begin{aligned}
 y &= ax^2 + bx + c. \\
 y &= 5(x - 2)^2 + 1 \\
 y &= 5(x - 2)(x - 2) + 1 \\
 y &= 5(x^2 - 4x + 4) + 1 \\
 y &= 5x^2 - 20x + 20 + 1 \\
 y &= 5x^2 - 20x + 21
 \end{aligned}$$

9.2.3 Collaboration: Writing a Quadratic Function Using a Vertex and Point From a Graph (continued)

3. Write the quadratic equation for the graph.



Vertex Form	$y = a(x + 1)^2 - 3$ $-15 = a(3 + 1)^2 - 3$ $-15 = a(4)^2 - 3$ $-15 + 3 = a(16)$ $-12 = a(16)$ $a = -\frac{3}{4}$ $y = -\frac{3}{4}(x + 1)^2 - 3$
Standard Form	$y = -\frac{3}{4}(x + 1)^2 - 3$ $y = -\frac{3}{4}(x + 1)(x + 1) - 3$ $y = -\frac{3}{4}(x^2 + 2x + 1) - 3$ $y = \left(-\frac{3}{4}x^2 - \frac{3}{2}x - \frac{3}{4}\right) - 3$ $y = -\frac{3}{4}x^2 - \frac{3}{2}x - \frac{15}{4}$



The conversion from vertex to standard may cause challenges for students who forget how to expand a binomial. Students who struggle should write out both binomial factors. Listen for students who have the misconception that $(x + 1)^2 = x^2 + 1$.



Students may have fraction operation deficits from prior grade levels. If these deficits exist, students may struggle with the standard form version of question 3.

9.2.4 Your Turn!

Component Overview: Students will work independently to practice writing a quadratic function in vertex and standard forms when given a graph, vertex, and a point on the graph.



9.2.4 Your Turn!

1. Complete the table for each quadratic function using the graph of the function.

Graph A	Vertex and Point	Vertex Form
	Vertex: (7, -6)	$y = a(x - 7)^2 - 6$ $3 = a(4 - 7)^2 - 6$ $3 = a(-3)^2 - 6$ $3 + 6 = a(9)$ $9 = a(9)$ $a = 1$ $y = (x - 7)^2 - 6$
	Point: (4, 3)	
		Standard Form $y = (x - 7)(x - 7) - 6$ $y = (x^2 - 14x + 49) - 6$ $y = x^2 - 14x + 43$
Graph B	Vertex and Point	Vertex Form
	Vertex: (-3, 5.5)	$y = a(x + 3)^2 + 5.5$ $-3.5 = a(0 + 3)^2 + 5.5$ $-3.5 = a(3)^2 + 5.5$ $-3.5 - 5.5 = a(9)$ $-9 = a(9)$ $a = -1$ $y = -(x + 3)^2 + 5.5$
	Point: (0, -3.5)	
		Standard Form $y = -x(+3)(x + 3) + 5.5$ $y = -(x^2 + 6x + 9) + 5.5$ $y = -x^2 - 6x - 9 + 5.5$ $y = -x^2 - 6x - 3.5$



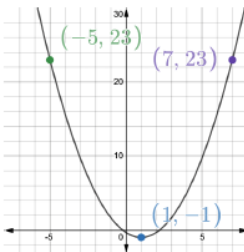
Help students connect the forms with the points on the graph. Students may have a disconnect between the vertex form and the vertex, or the a -value of both forms and whether the graph opens up or down. Finding the x -intercepts on the graph and noting that they're neither vertex nor standard form easily reveals the x -intercepts.



Use inquiry to promote synthesis and conceptual understanding.

- What is the axis of symmetry?
- Which form is easiest to use to determine the axis of symmetry?
- Draw the axis of symmetry on the graph. What is the y -intercept?
- Which form is easiest to use to determine the y -intercept? Where is it on the graph?

9.2.4 Your Turn! (continued)

Graph C	Vertex and Point	Vertex Form
	Vertex: (1, -1)	$y = a(x - 1)^2 - 1$ $23 = a(7 - 1)^2 - 1$ $23 = a(6)^2 - 1$ $23 + 1 = a(36)$ $24 = a(36)$ $a = \frac{2}{3}$ $y = \frac{2}{3}(x - 1)^2 - 1$
	Point: (7, 23)	
		Standard Form $y = \frac{2}{3}(x - 1)(x - 1) - 1$ $y = \frac{2}{3}(x^2 - 2x + 1) - 1$ $y = \frac{2}{3}x^2 - \frac{4}{3}x - \frac{1}{3}$



Preview Lesson 3 by engaging students in a discussion on the relationship between increasing x -values and decreasing y -values. When does each value increase or decrease? What does the vertex have to do with it?

9.2.5 Wrap-Up: Reflection

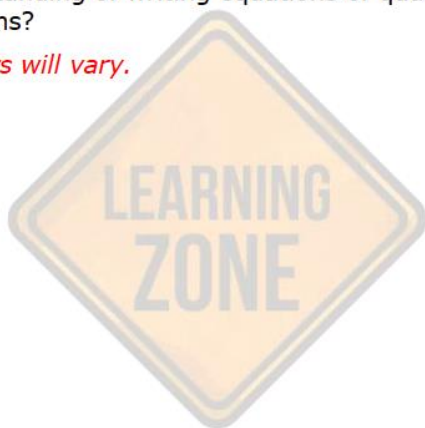
Component Overview: Students will complete a self-reflection on their level of confidence in writing equations of quadratic functions. Students should be able to articulate their reflection in their own language.



9.2.5 Wrap-Up: Reflection

1. How “in the zone” do you feel right now about your understanding of writing equations of quadratic functions?

Answers will vary.



Consider prompting students to use their response from Lesson 1 to build their response in Lesson 2. What is still “muddy” and what about this lesson helped clarify certain points of uncertainty? What do you still have questions about?